Model Comparison & Evaluation Report

Project: Aerial Object Classification & Detection (Bird vs. Drone)
Objective: Evaluate and compare Deep Learning architectures to determine the optimal model for real-time aerial surveillance deployment.

1. Executive Summary

This project evaluated three distinct deep learning approaches to solve the problem of distinguishing drones from birds in aerial imagery. The models evaluated were a **Custom CNN** (baseline classification), **ResNet50 Transfer Learning** (advanced classification), and **YOLOv8** (classification + object detection).

While the Custom CNN outperformed the heavier ResNet50 model in pure classification due to domain-specific feature learning, **YOLOv8** was ultimately selected as the champion model. YOLOv8 not only provided exceptional classification accuracy but also solved the critical real-world requirement of spatial localization (bounding boxes).

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2. Model Architectures & Performance

Model 1: Custom Convolutional Neural Network (CNN)

- **Approach:** Built from scratch. 4-block architecture with progressive filter sizes (32 to 256), Batch Normalization, and Dropout layers.
- **Input Size:** 224x224x3
- **Performance:**
  - **Accuracy:** 85.12%
  - **Precision:** 86.4%
  - **Recall:** 83.8%
  - **F1-Score:** 85.0%
- **Pros:** Highly optimized for our specific dataset. Lightweight and fast to train. Learned the specific edge-cases of aerial silhouettes well.
- **Cons:** Cannot locate the object in the image (no bounding boxes). Struggles slightly if multiple objects are in the same frame.

Model 2: Transfer Learning (ResNet50)

- **Approach:** Pre-trained on ImageNet. Base layers were frozen, and a custom dense classification head was added. Fine-tuned with a low learning rate (1e-4).

- **Input Size:** 224x224x3
- **Performance:**
    - **Accuracy:** 80.45%
    - **Precision:** 81.1%
    - **Recall:** 79.5%
    - **F1-Score:** 80.2%
- **Pros:** Requires less data to train from scratch. Excellent at recognizing complex textures.
- **Cons:** Suffered from **Domain Mismatch**. ImageNet is trained on large, everyday objects (dogs, cars, etc.). Our dataset consists of tiny, dark silhouettes against bright skies. The highly complex ResNet features caused slight overfitting, resulting in poorer performance than the simpler Custom CNN.

Model 3: YOLOv8 (Object Detection) - *Champion Model*

- **Approach:** YOLOv8 Nano architecture (yolov8n.pt). Trained on YOLO-formatted .txt labels to predict both class labels and spatial bounding boxes.
- **Input Size:** 416x416x3
- **Performance:**
    - **mAP50 (Mean Average Precision):** 92.5%
    - **Classification Precision:** 94.2%
    - **Recall:** 91.8%
    - **F1-Score:** 92.9%
- **Pros:** State-of-the-art inference speed. Solves the spatial localization problem. Can detect *multiple* objects (e.g., a flock of birds and a drone) in the exact same frame.
- **Cons:** Requires a more complex dataset format (bounding box annotations rather than just sorting images into folders).

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3. Quantitative Comparison Table

Metric	Model 1: Custom CNN	Model 2: ResNet50	Model 3: YOLOv8 (Nano)

Task Type	Image Classification	Image Classification	Object Detection
Input Image Size	224 x 224	224 x 224	416 x 416
Accuracy / mAP	85.12%	80.45%	**92.50%** (mAP50)
Precision	86.40%	81.10%	**94.20%**
Recall	83.80%	79.50%	**91.80%**
F1-Score	85.00%	80.20%	**92.90%**
Model Weight Size	~15 MB	~95 MB	**~6 MB**
Overfitting Risk	Low (EarlyStopping)	Medium	Very Low

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4. Final Selection & Business Impact

Selected Model: YOLOv8 (Model 3)

Justification for Business Impact:
For real-world use cases such as Airport Bird-Strike Prevention and Restricted Airspace Security, simply classifying an image as "Bird" or "Drone" is insufficient. Security personnel and air traffic controllers must know the exact location, quantity, and trajectory of the objects in the camera frame.

I selected YOLOv8 as the final model for the Streamlit deployment because it perfectly balances high-speed inference with spatial accuracy. Furthermore, because we utilized the YOLOv8 Nano variant, the model size is incredibly lightweight (~6 MB), allowing it to run smoothly on edge devices or web applications without requiring expensive cloud GPUs during deployment.

The high **Recall (91.8%)** ensures that dangerous False Negatives (e.g., failing to detect a drone near a runway) are kept to an absolute minimum, successfully fulfilling the safety and security objectives of the project.